MATH 101 Unit 2: Functions and Their Graphs - Practice 3

Version 202610122020.39 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. (a) Choose the pair giving h(x) = f(g(x)). (b) Form F(G(x)) and evaluate it at the supplied input.

$h(x) = (4x + 3)^2 + 5$

A. $f(x) = x^2 + 5,\ g(x) = 4x + 3$

B. $f(x) = 4x + 3,\ g(x) = x^2 + 5$

C. $f(x) = x^2 + 3,\ g(x) = 4x + 5$

D. $f(x) = 4x^2 + 5,\ g(x) = x + 3$

(b) $F(x) = x^2,\ G(x) = x - 4,\ x_0 = 5$

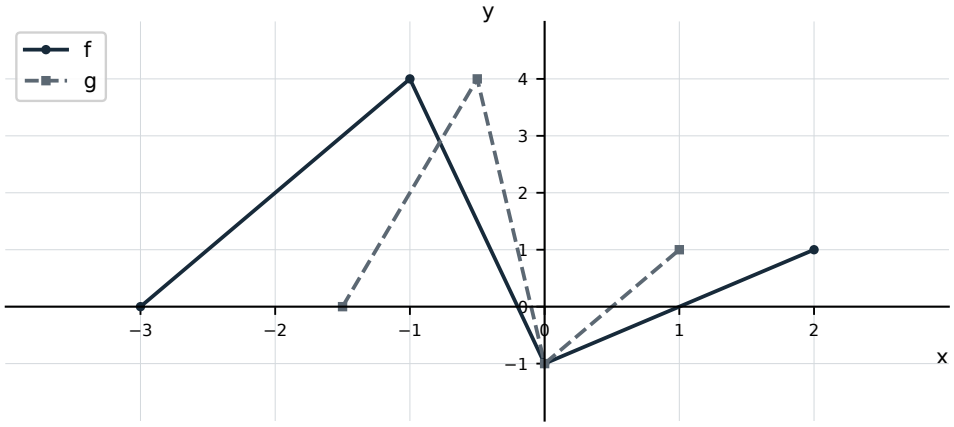
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x/2)$

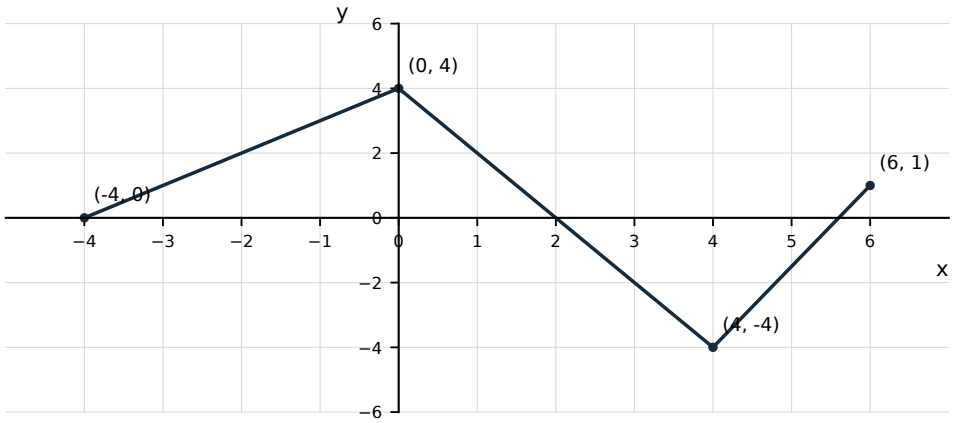
B. $g(x) = 2f(x)$

C. $g(x) = f(2x)$

D. $g(x) = f(x - 2)$



3. From the graph, find (a) average rate from x = -4 to x = 6; (b) increasing and decreasing intervals; (c) the two local extrema (ordered pairs).



4. (a) Find the linear function through these points, with domain all real numbers. (b) Find its inverse and give the inverse domain and range.

$(1,\ 5);\ (3,\ 13)$

5. For this function, find (a) f(-3), (b) domain, and (c) range.

$f(x) = 4|x - 3| - 4$

Student answer sheet

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Question 1

(a) Choice

8 pts

(b) Composition

8 pts

(b) Value

4 pts

Question 2

Choice

20 pts

Question 3

(a) Rate

6 pts

(b) Increasing

4 pts

(b) Decreasing

4 pts

(c) Local max

3 pts

(c) Local min

3 pts

Student answer sheet

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Question 4

(a) Function

10 pts

(b) Inverse

6 pts

(b) Domain

2 pts

(b) Range

2 pts

Question 5

(a) Value

6 pts

(b) Domain

7 pts

(c) Range

7 pts

Answer key

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Question 1

(a) Choice	A	8 pts
(b) Composition	$x^2 - 8x + 16$	8 pts
(b) Value	1	4 pts

Question 2

Choice	C	20 pts

Question 3

(a) Rate	$\frac{1}{10}$	6 pts
(b) Increasing	$(-4, 0) \cup (4, 6)$	4 pts
(b) Decreasing	$(0, 4)$	4 pts
(c) Local max	$(0,\ 4)$	3 pts
(c) Local min	$(4,\ -4)$	3 pts

Answer key

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Question 4

(a) Function	$f(x) = 4x + 1$	10 pts
(b) Inverse	$f^{-1}(x) = \frac{x}{4} - \frac{1}{4}$	6 pts
(b) Domain	$(-\infty, \infty)$	2 pts
(b) Range	$(-\infty, \infty)$	2 pts

Question 5

(a) Value	20	6 pts
(b) Domain	$(-\infty, \infty)$	7 pts
(c) Range	$[-4, \infty)$	7 pts